

What the correlations between neurons say about each neuron

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Abstract

Record many neurons while they watch the same stimulus, then discard the stimulus. What remains is how the neurons relate to each other: a correlation matrix. We ask whether that matrix alone says what each neuron represents. A transformer that sees only the correlation matrix of 512 sampled neurons, with no labels and no reference population in its input, decodes the preferred orientation of neurons of mouse visual cortex to a mean error of 25° (uninformed: 45° ; best constant guess given the label distribution: 41° ; 28° when every orientation is weighted equally) and their receptive-field position at $R^2 = 0.23$, from in-vivo responses in the MICrONS volume; on the digital-twin substrate it reaches 20° and 0.48. Put simply, asked whether a neuron prefers horizontal or vertical, the decoder is right for 79% of neurons in vivo and 87% on the twin, against 56% for the majority answer; what it does not resolve is the obliques. The readout is possible because the relations between orientation classes are close to circulant, which fixes the class structure up to rotation and reflection; the absolute frame is fixed by a departure from that symmetry which persists when every orientation is equally represented in each population and when training and test relations come from disjoint stimuli, and which follows each area's dominant orientation: the neurons that prefer it co-fluctuate most strongly, and a model population with only that property recovers the absolute frame. Across 33 mice of the Allen Brain Observatory, with relations computed from responses to drifting gratings, orientation transfers to unseen animals only when a sampled population mixes animals, so that its matrix contains between-animal correlations; populations drawn from one animal carry only a weak signal, 2° to 3° below chance, a limit of the training diversity one animal provides rather than of its circuit, since single-scan populations in MICrONS are decoded to 31° to 33° . For receptive fields, what crosses animals is where each animal's imaged patch looks on the screen, and not the layout of neurons within it. Capacity helps where labels are plentiful and stops helping where they are scarce.

1 Introduction

What a neuron in visual cortex represents is established by relating its activity to the stimulus: a neuron prefers vertical gratings, or responds to the upper left of the screen. This paper asks whether the same facts are contained in a description that mentions no stimulus at all, the correlations between the neuron and the other neurons recorded with it. Take a population that watched the same movie, compute the correlation of every pair of neurons, and discard the movie. Does the resulting matrix determine which neuron prefers which orientation, and where each neuron's receptive field lies?

The idea that content can be read from relations is old. Shepard and Chipman proposed that internal representations need not resemble what they stand for as long as the similarities among them mirror the similarities among the things represented, a second-order isomorphism [1]; Edelman made the case that representation is representation of similarities [2]. Representational similarity analysis turned the idea into a method that compares brains, models and species

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through their similarity structures rather than their coordinates [3, 4], hyperalignment builds a shared representational space across subjects from those structures [5], and the same principle underlies current work on aligning representations across networks and individuals [6, 7]. In neuroscience, the correlation structure of spontaneous activity was shown to carry the orientation map of visual cortex without any stimulus [8, 9] and, more generally, to match the statistics of evoked activity [10]; functionally similar neurons connect preferentially [11, 12]; and population dynamics are preserved across animals performing the same behaviour [13]. Structuralist accounts of experience make the general claim that what a state represents, or how it feels, is fixed by its position in a space of relations [14], with Kleiner and Ludwig [15] giving a formal definition of such structures; Lässig [16] formulates the corresponding question for neural networks and gives the decoding framework we use here.

The spontaneous-activity results are the closest precedent, and they set the premise: a neuron’s correlations with the population contain its tuning even in the dark. Tsodyks et al. [8] showed it for single neurons by averaging the imaged population activity at the times a neuron fired and finding the evoked map of its preferred orientation. Reading the tuning out of that pattern, however, required the evoked map, a stimulus-labelled reference: correlations sort neurons by orientation and place the groups in order around the orientation circle, but on their own they say nothing about which group is vertical. The same holds for representational similarity analysis and hyperalignment, whose structures are over conditions or subjects and whose interpretation goes through a known stimulus set. Our question is one level down and one step further: whether the relations among *neurons*, with the stimulus that produced them discarded and no labelled neuron or map to compare with, fix the content of each neuron, and whether that content is readable at the scale of thousands of individually identified neurons and across animals. The closest neighbours in machine learning for neural data read properties of a neuron from population activity: NeuPRINT learns a time-invariant embedding of each neuron from the dynamics of its population and decodes cell type from it [17], NuCLR does the same with a contrastive objective and a permutation-equivariant transformer, and transfers cell type and brain region to unseen animals [18], and POYO trains one decoder across sessions and animals by tokenising spikes with learned unit embeddings [19]. These models see the activity itself; the present decoder sees only the correlation matrix, so the question it answers is what the relations alone determine, and the content it targets is stimulus tuning rather than cell class. In practice the relational situation is often the one at hand: recordings under natural behaviour have no controlled stimulus, and neurons from different sessions or animals share no stimulus-aligned coordinate system, but they do share the structure of their correlations. In principle, the answer says how much of a neuron’s content is fixed by its position in the network of relations among neurons.

We answer it with a decoder. The input is the correlation matrix of a sampled population of a few hundred neurons, one row per neuron, and nothing else: no labels and no reference neurons. The output is a preferred orientation and a receptive-field position for every neuron in the population. The decoder is trained on populations whose labels are known and scored on neurons it never saw. We do this at two scales. Within one brain, using the MICrONS release [12, 20], 12,894 neurons from a cubic millimetre of mouse visual cortex with in-vivo responses to a shared movie, a fitted digital-twin model [21], and labels for orientation and receptive field (Fig. 1): the decoder trained on half the neurons reads the other half. Across brains, using the Allen Brain Observatory [22], 33 mice recorded under identical stimuli: the decoder trained on some animals reads neurons of animals it never saw.

The results come with an explanation. Averaging the correlations by orientation class shows that the relations between classes are almost circulant, which fixes the classes only up to a rotation or reflection of the orientation circle; a small, consistent departure from that symmetry is what pins the absolute frame and makes the readout possible.

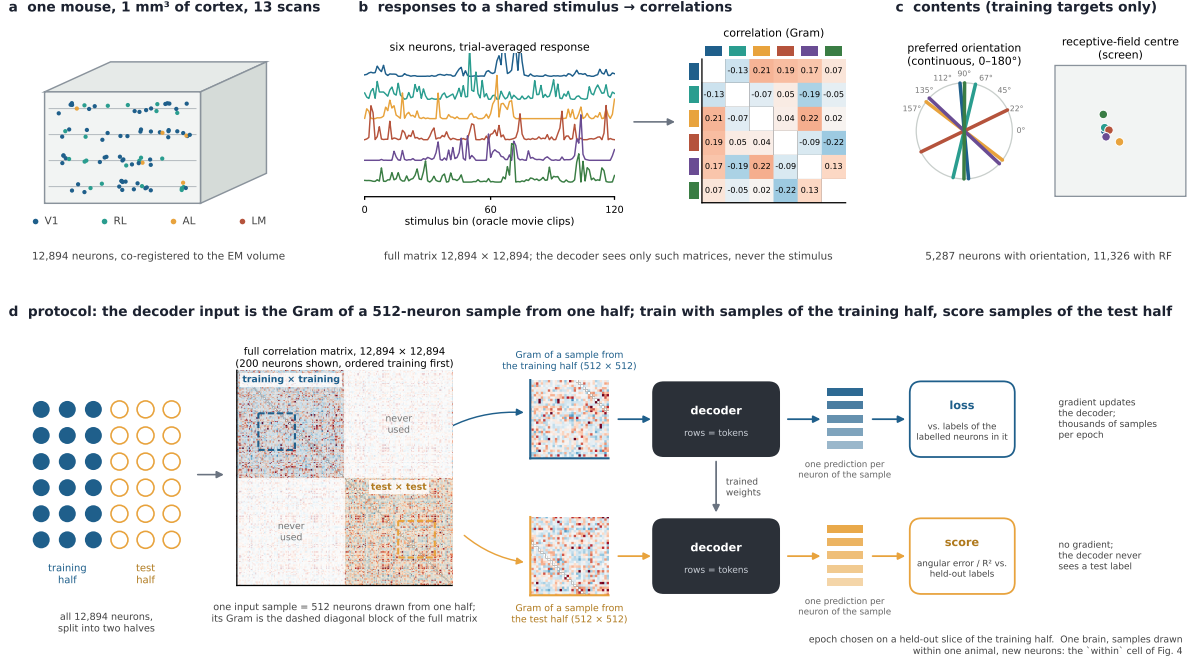


Figure 1: How the MICrONS task is built. (a) One mouse, 13 two-photon scans of a cubic millimetre of visual cortex (areas V1, RL, AL, LM), 12,894 neurons co-registered to the electron-microscopy volume. (b) The relational substrate. Each neuron is its trial-averaged response vector to a stimulus all neurons saw (in vivo: 120 bins of the oracle natural-movie clips; digital twin: 4,999 bins compressed to 512 principal components); shown are six real neurons and their correlation matrix. The oracle clips were shown in all 13 scans, so the matrix is defined across scans. The decoder only ever sees such correlations, never the stimulus. (c) The contents of the same six neurons: preferred orientation (in vivo, a continuous angle, 5,287 neurons with $gOSI \geq 0.25$) and receptive-field centre (digital-twin fit, 11,326 neurons). These are training targets only. (d) The protocol, with two terms kept apart. The *halves*: all neurons are split at random into a training half and a test half, and the full 12,894 × 12,894 correlation matrix, ordered training-first (a real 200-neuron excerpt is shown), is used only in its training×training and test×test blocks. A *sample*: 512 neurons drawn at random from one half; its Gram is the corresponding diagonal sub-block (dashed), and that 512 × 512 matrix is the decoder’s entire input, one row per neuron, one prediction per row. Training draws thousands of samples per epoch from the training half; their predictions meet the labels of the labelled neurons in the sample in a loss that updates the decoder. Scoring draws samples from the test half and compares predictions with held-out labels; no gradient. Labels never enter the input on either path, and the halves share no neuron. The epoch is chosen on a held-out slice of the training half.

2 Methods

2.1 Data

MICrONS. The release of Ding et al. [12] contains 12,894 neurons co-registered to the electron-microscopy volume, from 13 two-photon scans across V1, RL, AL and LM. Each neuron has a trial-averaged in-vivo response to the oracle natural-movie clips (120 bins), shown in every scan; a digital-twin response to a shared movie (4,999 bins, which we compress to 512 principal components; the Gram is preserved to $r > 0.999$); an in-vivo preferred orientation with a global orientation selectivity index (gOSI); and a receptive-field centre from the twin (spike-triggered-average fit, screen coordinates in $[-1, 1]$). We use orientation, as a continuous angle, for the 5,287 neurons with $\text{gOSI} \geq 0.25$, and receptive-field position for the 11,326 neurons whose twin test correlation is ≥ 0.2 . The orientation labels are in-vivo measurements on both substrates; the receptive-field labels come from the twin’s fit on both substrates, so every receptive-field result, in vivo included, scores relations against model-derived positions (Section 4). Same-scan pairs correlate about 50% more strongly than cross-scan pairs at matched receptive-field distance; the content of the correlations is the same across scans, and all populations mix scans. Only 2% of the receptive-field label variance lies between scans and 9% between areas, so a decoder that read scan or area identity from the Gram could reach at most that much R^2 .

Allen. From Visual Coding 2P [22] we take the 36 VISp excitatory containers with at least 150 orientation-labelled cells: 33 mice, 9,281 cells, of which 5,529 carry an orientation label; unlabelled cells stay in the populations and receive no loss, so a held-out-animal split has 6,706 training cells and 2,575 test cells in total. Relations come from session A, either responses to natural movies shown to every mouse (4,500 bins) or the 40 blank-subtracted condition means of drifting gratings (8 directions $\times$ 5 temporal frequencies). Orientation labels come from static gratings in session B (six orientations 30° apart, $\text{gOSI} \geq 0.25$, $p < 0.05$), so labels and relations use different stimuli. Receptive-field centres come from locally sparse noise in session C (1,504 cells with a significant fit, 23 to 112 per mouse).

2.2 Relations

Each neuron’s response vector is centred and normalised; the relation between two neurons is the cosine of their vectors, a signal correlation. For a population of n neurons the Gram is the $n \times n$ matrix of these values, standardised by the global off-diagonal mean and standard deviation, with the diagonal zeroed; we keep the word Gram for this standardised correlation matrix throughout.

2.3 Decoder

Each row of a population’s Gram, together with three row statistics (the mean, standard deviation and third moment of the row), is projected to a token; L pre-norm transformer blocks [23] with H -head attention update the tokens, the Gram entering the attention logits as an additive bias; a linear head reads out, per token, the unit vector $(\cos 2\theta, \sin 2\theta)$ of the preferred orientation θ or an (x, y) receptive-field position. The loss is the mean squared error on the labelled tokens only; unlabelled neurons stay in the population and receive no loss. The standard configuration is width 512, 8 layers and 17M parameters with populations of 512 neurons on MICrONS, and width 256, 4 layers and 2.2M parameters with populations of 256 neurons on Allen; other sizes are reported in the appendix. Training draws thousands of random populations per epoch.

2.4 Protocol

Split. MICrONS neurons are halved at random. Training populations are sampled inside the training half and test populations inside the test half, so no test neuron ever appears in a training

population (Fig. 1d). A slice of the training half is held out to choose the epoch; the test half never influences selection.

Regimes (Allen). Two independent choices define a run (Fig. 4). The split decides where test neurons come from: the training animals, each halved into training and test neurons, or held-out animals. The population decides what a sample is: neurons drawn from one animal, so that the Gram is a within-circuit matrix, or from several animals, so that most entries are between-animal correlations. We call the four cells **within**, **pooledwithin**, **cross** and **pooledcross**. Held-out regimes select the epoch on further held-out mice. Every held-out cell is run on three mouse splits and every training-animal cell on three neuron splits.

Metrics. Orientation is scored by the mean absolute angular error between the decoded angle, $\frac{1}{2} \text{atan2}$ of the output, and the label, taken modulo 180° , so it lies between 0° and 90° . A decoder that carries no information produces errors spread evenly over that range, whatever it outputs, so chance is 45° . Allen labels sit on a 30° grid: a decoder that recovered every true preference exactly would still disagree with those labels by 7.5° on average, which is the floor to read Allen errors against. A frame check reports the error after the best single rotation or reflection of all decoded angles; if it is lower than the raw error, the decoder recovered the structure but not the frame. Receptive fields are scored by R^2 against a baseline of 0.

2.5 Symmetry analysis

To expose the symmetry of the relations we average the Gram into a $K \times K$ class-Gram, the mean relation between neurons of two orientation classes, project it onto its circulant part, and ask which relabellings of the classes a held-out half’s class-Gram is compatible with (all $K!$ relabellings, Frobenius distance). A relabelling that differs from the truth by a rotation or a reflection of the orientation circle is an element of the dihedral group D_K ; the analysis reports whether the truth is singled out absolutely or only modulo D_K .

3 Results

3.1 Per-neuron content from the correlation matrix alone

On MICrONS the decoder reads orientation and receptive-field position from the Gram of 512 neurons of the test half (Fig. 2, Table 1). The twin substrate is cleaner than the in-vivo one, since the twin removes trial noise, and receptive fields carry more than orientation. The predictions are in the true frame: the frame check lowers the error by less than 0.1° in every run, so the decoder places each neuron at its absolute orientation rather than recovering the structure up to a rotation. About half of the neurons are decoded to within 15° (0.46 in vivo, 0.54 on the twin). The error is not spread evenly over the orientation circle: the decoder resolves the cardinal axes and little else (Appendix B, Fig. A5). Asked only which cardinal axis a neuron is closer to, it is right for 87% of all neurons on the twin and 79% in vivo (chance 56%), and for the 56% of neurons whose preference lies within 15° of an axis, for 94% and 86%. Sorted into four classes at 0° , 45° , 90° and 135° , 66% (twin) and 59% (in vivo) land in the right one against a majority class of 40%; the misses are almost all obliques. Both readouts are properties of individual neurons rather than of the scan they were recorded in: centring predictions and labels per scan leaves the receptive-field R^2 unchanged and the orientation error within 0.2° (Appendix B). Test Grams built from stimulus bins the decoder never trained on cost 1° on the twin and 9° in vivo, where a Gram over 60 movie bins is partly specific to those bins (Appendix B). The label distribution sets a second baseline: the best constant guess scores 41.3° , and the error averaged over eight equally weighted orientation bins is 24.5° on the twin and 27.9° in vivo (Table 1).

Table 1: MICrONS, per-neuron decoding on the test half. Relations only: the 17M decoder on the Gram of 512-neuron populations, mean $\pm$ s.d. over 3 seeds. Balanced: the error averaged over eight equally weighted true-orientation bins (neuron split 2). Stimulus-known linear readout: a linear map from each neuron’s own movie response vector to its content, a reference that uses the stimulus alignment the relational decoder discards, not a label-noise ceiling (Appendix B). Chance: an uninformed decoder (45°) and the best constant orientation given the label distribution (41.3°).

content	substrate	relations only	balanced over orientations	stimulus-known linear readout	chance / best constant
orientation (angular error)	in vivo	$25.4^\circ \pm 0.5^\circ$	27.9°	20.2°	45° / 41.3°
orientation (angular error)	twin	$20.2^\circ \pm 0.4^\circ$	24.5°	15.8°	45° / 41.3°
receptive field (R^2)	in vivo	0.234 ± 0.007	–	0.34	0
receptive field (R^2)	twin	0.480 ± 0.010	–	0.68	0

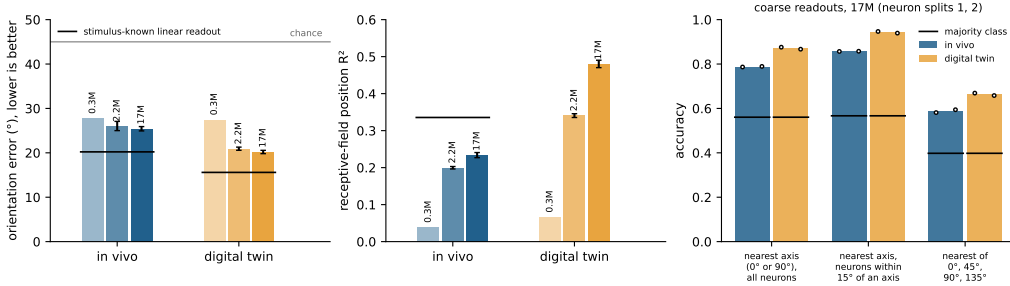


Figure 2: MICrONS: per-neuron content decoded from the population correlation matrix alone. Left: mean angular error of the decoded preferred orientation on held-out neurons of the same animal, for decoders of 0.3M, 2.2M and 17M parameters; error bars are the s.d. over 3 seeds; grey line: the 45° error of an uninformed decoder; black marks: the stimulus-known linear readout of each neuron’s own responses. Middle: R^2 of the decoded receptive-field centre, same marks. Right: coarse readouts of the same decoder on neuron splits 1 and 2: whether the decoded angle falls on the correct cardinal axis, for all neurons and for the neurons whose preference lies within 15° of an axis, and whether it falls in the correct one of four classes at 0° , 45° , 90° and 135° ; black ticks: majority class. The decoder sees only the standardised correlation matrix of 512 sampled neurons.

3.2 Why the absolute frame is recoverable

The class-Gram of orientation is close to circulant (Fig. 3): neighbouring orientations correlate, orthogonal ones anti-correlate, and the pattern repeats around the circle. A circulant matrix is unchanged by every rotation and reflection of the orientation circle, so that part of the relations fixes the classes only up to the 16 elements of D_8 and cannot say which class is 0° . Whatever fixes the frame must be a departure from circulant structure, at the class level or finer. At the class level that residual holds 29% of the class-Gram variance in vivo and 19% on the twin, and it is signal: it correlates at 0.94 between random halves of the neurons, at 0.84 and 0.99 between disjoint halves of the stimulus bins, and at 0.99 with the residual computed on class-balanced subsamples, so it is neither a finite-stimulus artefact nor a consequence of the uneven number of neurons per orientation. Its form is a cardinal anisotropy, stronger relations among neurons near the cardinal axes, whose shape differs between areas (residual correlation V1–RL 0.77, V1–AL 0.31).

Five observations locate the structure in the circulant part and narrow down what fixes the frame. First, the decoder recovers the absolute frame (Section 3.1), while predictors that use only the distribution of labels, the best constant orientation or each scan’s or area’s mean, reach 39° to 41° . Second, a decoder trained with no frame in its labels, each population’s targets rotated at random and the loss taken after the best alignment of predictions to targets, places neurons to 19.9° on the twin and 25.5° in vivo after a single global alignment, the same as the standard decoder with the frame (20.2° , 25.4°): the circulant part carries the whole per-neuron structure, and the frame comes at no cost. That decoder’s raw error, 25° and 31° against 45° for an arbitrary frame, shows it anchors a consistent frame on the data even when nothing in the loss asks for one. Third, the frame is not the uneven number of neurons per orientation. With every training and test population drawn with equal numbers of neurons from each of the eight orientation classes, the decoder reaches 22.0° on the twin and 26.6° in vivo (two seeds each), with no gap between raw and aligned error, and it is better than the standard decoder scored with the same equal weighting (24.5° , 27.9°). Nor is the frame the stimulus sample: it survives training and test relations built from disjoint halves of the stimulus (Appendix B). Fourth, the errors have the shape of the symmetry breaking, 11° to 12° near the cardinal axes and 37° to 40° at the obliques on the twin, and reweighting the loss so that obliques count as much as cardinals leaves them unchanged (Appendix B); the oblique ordering is not in the relations at single-neuron resolution. Fifth, a decoder trained on V1 transfers to RL, whose anisotropy resembles V1’s, with a 1.6° frame gap, and to AL, whose anisotropy differs, with a raw error of 58° that falls to 32° after one global rotation of 78° : it carries the structure and the wrong frame, and the rotation it needs is close to the 90° that separates the two areas’ dominant orientations (28% of V1 neurons prefer 90° , 32% of AL neurons 0°). The dominant orientation of an area is also its most strongly co-correlated one: the mean correlation between two neurons of the same class peaks at 90° in V1 (0.071 against 0.047 to 0.063 for the other classes on the twin, 0.095 against 0.046 to 0.082 in vivo) and at 0° in AL (0.204 against 0.06 to 0.19), and these values are unchanged on class-balanced subsamples, so the signature is in the correlations, not in the counts. Taken together, the frame follows the area’s over-represented orientation through the stronger co-fluctuation of the neurons that prefer it, a property each neuron’s Gram row carries whatever the composition of the sampled population. We could not isolate the anchor by subtraction: removing the class-level residual from every Gram with the true classes injects the classes into the Gram (a control that removes the whole class structure improves the score), so the frame is located by exclusion and by the cross-area pattern, not by a direct ablation of the candidate. A synthetic population shows the two halves of the logic separately and gives the candidate anchor a direct test (Appendix B, Table 5): model neurons with von Mises tuning, a movie-like stimulus that covers orientation evenly, and the same pipeline give a class-Gram circulant to within noise, and the decoder trained on it recovers the ordering, 30° after one global alignment, but no frame, 54° raw. Sharper cardinal tuning and a cardinal excess of neurons do not fix the frame at any

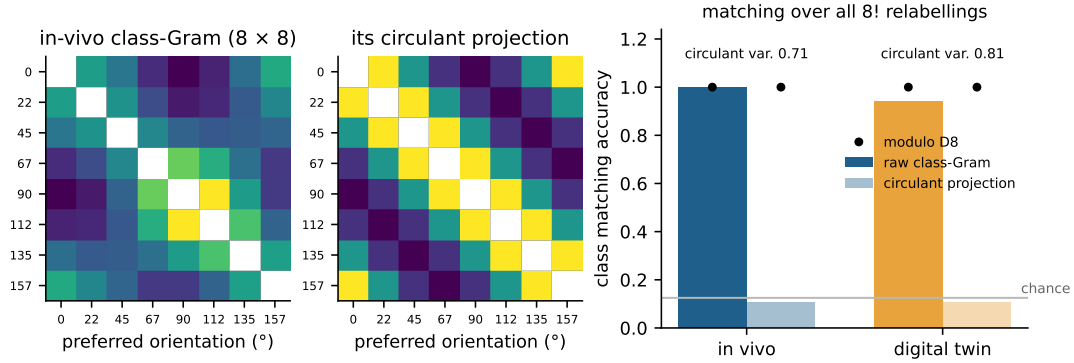


Figure 3: What fixes the absolute orientation frame. Left: mean correlation between orientation classes in vivo (diagonal masked). Middle: its circulant projection. Right: matching a held-out half’s class-Gram to the reference over all 8! relabellings. With the raw matrix the true labelling wins. After the circulant projection, absolute accuracy collapses to chance while accuracy modulo D_8 (dots) stays at 1.0, as it must, since a circulant matrix is invariant under D_8 , and the bars sit at or slightly below the chance line because the matching then picks one of the 16 equivalent labellings at random; the panel illustrates what the residual has to supply.

strength tested; an excess of cardinal frames in the stimulus does in most seeds. Adding only the anisotropy the data show, stronger co-fluctuation among like-tuned neurons peaked at one orientation, with uniform counts and an even stimulus, fixes the frame: 9° raw at a same-class correlation raised 1.55-fold at 90° (four of five seeds), 3° at 2.1-fold, and in one of two seeds at 1.2-fold, the factor of the twin V1 data. The cortical anchor is not stimulus-inherited, since it reproduces across disjoint halves of the stimulus and differs between areas that saw the same movie.

3.3 Orientation transfers between animals through mixed populations

With natural-movie relations, orientation is nearly absent from Allen correlations: same-orientation pairs correlate 0.005 more than orthogonal pairs, against 0.09 in MICrONS, and every test is at chance; the Allen results below therefore use grating relations, a designed set of 40 conditions over an orientation grid, and do not replicate on naturalistic relations. With grating relations the signal is present (0.05 within a mouse, 0.044 across mice), and the decoder’s four cells separate (Fig. 5, Table 2). Single-animal populations carry a weak but consistent signal, 2° to 3° below chance in all six runs, whether or not the animal was seen in training. Mixed populations are 7° to 10° below chance in both rows, at the same level, so transfer to a new brain has no cost detectable at three splits; as on MICrONS the readout is cardinal: over the three splits the decoder names the right cardinal axis for 62% of all cells and 68% of the cells labelled 0° or 90° (chance 51% and 52%), and the cells labelled 120° or 150° are decoded at chance (Fig. A5); what matters is whether the Gram contains between-animal correlations. Those exist only because all mice saw the same 40 conditions, and they measure how similar two tuning profiles are. The decoder still never sees which condition is which, so the absolute frame must again come from the population’s relational structure. In this dataset a neuron’s orientation is readable from where it sits in a correlation geometry that spans animals. From its relations inside its own circuit it is barely readable here, and two controls say why. The stimulus-known reference (Table 2, last row) shows the neurons are not the problem: with each neuron’s own 40-condition response vector as its token, the same decoder on the same single-animal populations reaches 33° to 35° , the same as with mixed populations. The training pool is: mixed populations drawn from a pool of only 280 or 560 training neurons, the size of two to four animals, score 40.5° and 40.7° , near chance, against 36.0° from the full pool of 6,706 at the same population size of 128. A single animal contributes about 140 training cells, so the decoder that sees only its within-animal

Table 2: Allen, mean angular error of the decoded orientation in the four regimes (2.2M decoder, 256-neuron populations; 128 for single-animal populations), three splits per cell (of neurons within each animal for the top row, of animals for the second row). Last row: the same decoder on held-out animals with each neuron’s 40-condition response vector as its token instead of its Gram row, the stimulus-known reference. Chance 45° ; label grid 7.5° .

test neurons from	single-animal populations	mixed populations
the training animals	42.5° / 42.9° / 42.1°	35.3° / 37.2° / 36.1°
held-out animals	43.1° / 43.1° / 42.7°	35.2° / 37.0° / 38.4°
held-out animals, stimulus-known reference	32.8° / 33.6° / 34.9°	32.4° / 33.7° / 34.3°

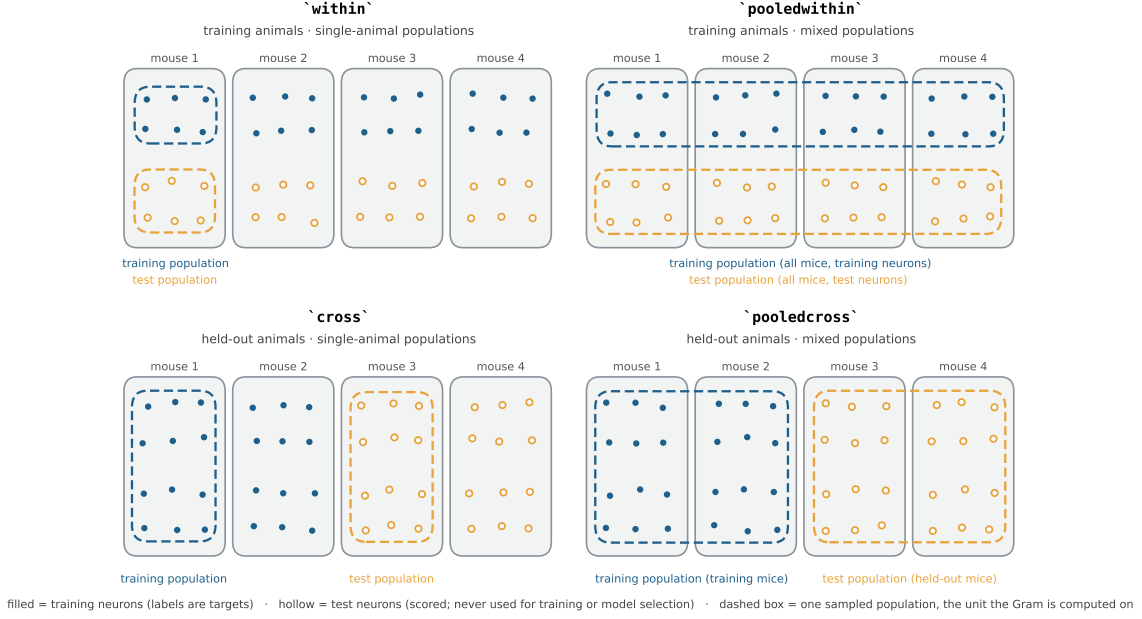


Figure 4: The four Allen decoder regimes. Rows: test neurons from the training animals (each animal’s cells halved) or from held-out animals. Columns: each sampled population drawn from one animal or from several. Filled dots are training neurons, hollow dots test neurons, dashed boxes one sampled population each. Training populations come only from training neurons and test populations only from test neurons in every regime.

correlations has too little diversity to learn from, whatever those correlations contain. The same limit reproduces on MICrONS: a decoder trained on 140 neurons of one scan and tested on the other scans scores 38° to 44° on the twin and 37° to 39° in vivo (three scans; Appendix B). On MICrONS, where one animal supplies thousands of training neurons, populations drawn from a single scan are decoded to 30.7° on the twin and 32.6° in vivo (128 neurons; Appendix B), against 22° for mixed-scan populations of the same size: relations inside one circuit do carry orientation, more weakly than relations that span circuits.

3.4 Receptive fields: what crosses animals is the animal’s screen position

Correlation falls with receptive-field distance within a mouse (0.127 at $< 5^\circ$ to 0.068 beyond 40°) and across mice (0.049 to 0.011). Decoding absolute screen coordinates on held-out mice gives R^2 that swings with the split, from 0.29 to -0.45 for single-animal populations and from 0.21 to -0.25 for mixed ones. Decomposing the saved predictions explains the swing (Fig. 6). The correlation between predicted and true mean position of each held-out mouse, pooled over 27 mouse-level predictions, is 0.36 to 0.71 (permutation $p \leq 0.03$). The correlation between predicted and true position of a neuron relative to its mouse-mates is -0.01 to 0.11 over 1,259

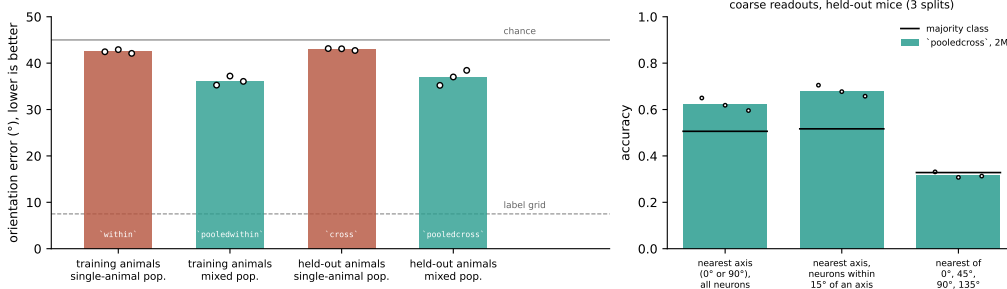


Figure 5: Allen, 33 mice, grating relations: orientation is readable only from populations that mix animals. Left: mean angular error; bars: mean over splits; points: the three splits; solid line: 45° chance; dashed line: the 7.5° label grid. Single-animal populations are 2° to 3° below chance whether the animal was seen in training or not; mixed populations are 7° to 10° below it in both rows. Right: coarse readouts of the mixed-population decoder on the held-out mice (three splits): correct cardinal axis for all cells and for the cells labelled 0° or 90°; the third bar, nearest of four classes 45° apart, is ill-defined for labels on a 30° grid and is shown only for comparison with MICrONS; black ticks: majority class.

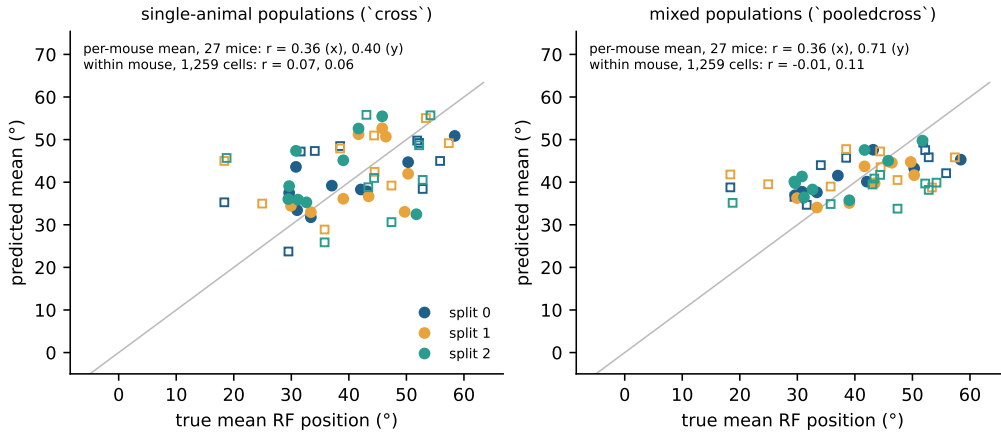


Figure 6: Allen receptive fields: what crosses animals is each animal's screen position. Predicted versus true mean receptive-field position of each held-out mouse (27 points over 3 splits; x squares, y dots), from decoders trained on single-animal (left) or mixed (right) populations. The per-mouse mean is recovered; the position of a neuron relative to its mouse-mates is not.

cells. The decoder recovers where an animal's imaged patch looks, a property of the population, and not which neuron within it looks where. R^2 on absolute coordinates is then dominated by whether the between-mouse spread of a particular test set is reproduced at the right scale.

The within-mouse layout is present in the correlations, since same-mouse correlation falls with distance. It is not learned because each mouse contributes 23 to 112 labelled cells. MICrONS, with 11,326 labels in one animal, is where the same decoder reaches neuron-level receptive fields at $R^2 = 0.48$.

3.5 What moves the numbers

Capacity pays where labels are plentiful and stops where they are scarce (Fig. A1). Twin receptive fields go from 0.34 to 0.48 to 0.51 R^2 between 2.2M, 17M and 57M parameters. Orientation error is flat from 2.2M upward on both MICrONS substrates (20.9°, 20.2°, 19.9° on the twin at 2.2M, 17M and 57M) and on Allen (35.2°, 36.5°, 36.7°); only the 0.3M model is worse (27.5° and 27.8° on MICrONS). Every model above 2M memorises its training neurons within 10 to 20 epochs and is early-stopped on the held-out selection slice (Fig. A3). Population size beyond 256 to 512

neurons changes orientation error by at most 2° in either direction, while twin receptive fields keep gaining slowly (Fig. A2). Recomputing each training population's Gram from a random subset of the stimulus conditions, a label-free augmentation, gives no consistent gain on the Allen error (35.2° , 33.5° , 37.7° and 38.0° with 100%, 75%, 50% and 30% of the conditions at 2.2M; 36.5° , 34.3° and 35.6° with 100%, 85% and 50% at 17M; single runs, against a split-to-split spread of 3°) and is not used. Split-to-split variance on held-out mice exceeds seed variance (Fig. A4).

4 Discussion

The correlation matrix of a few hundred cortical neurons, with the stimulus unknown and no neuron labelled, determines each neuron’s preferred orientation to a mean error of 25° in vivo and 20° on the digital twin, against 45° for an uninformed decoder, and its receptive-field position to $R^2 = 0.23$ in vivo and 0.48 on the digital twin. The mechanism is the one the symmetry analysis exposes: the relations fix the class structure up to the symmetries of the content, and a departure from that symmetry fixes the frame. That departure is a property of the correlations themselves: it is not the uneven number of neurons per orientation, not the particular stimulus sample, and it transfers between areas whose anisotropy matches. It follows each area’s over-represented orientation, whose neurons co-fluctuate most strongly, and it does so whether or not the sampled population is imbalanced; a direct ablation of that feature in the data is not available, but a model population with that co-fluctuation and nothing else recovers the absolute frame, while models with sharper or more numerous cardinal neurons do not.

Across animals the picture splits by content. Orientation lives in the relations between tuning profiles, which exist between any two neurons that saw the same conditions, in the same brain or not; from the within-circuit correlations of a single animal it is readable only weakly, at 31° to 33° on MICrONS, and on Allen not at all, where a single animal’s 140 training cells are too few to learn from. Receptive-field position is carried at the level of an animal’s patch, and the neuron-level layout needs more labels per animal than these recordings provide. Both statements are about what correlations under a shared stimulus contain, and both are measured under a protocol in which the test neurons, and in the held-out regimes the test animals, never touch training or model selection.

Three limits. MICrONS is one animal, and its training and test halves share recording sessions; the Allen held-out-animal cells are the cleaner generalisation test, and their variance across mouse splits is large. Orientation labels are noisy in both datasets (in vivo and twin labels agree on 70% of MICrONS neurons at 8 classes; static and drifting gratings on 45 to 67% of Allen cells), and the Allen labels are quantised to 30° , which caps every labelled and label-free number alike. The twin substrate is a model fitted to the same recordings, so the in-vivo results carry the claim about cortex and the twin results show what cleaner correlations would give; the receptive-field labels themselves come from the twin, so the twin receptive-field result compares a model’s correlations with the same model’s receptive fields, while the orientation labels are in-vivo measurements throughout.

The symmetry result is an instance of a general expectation: a representation that is equivariant to a group has a group-fixed part, identical across individuals up to relabelling, and an individual remainder [24]. The matching-over-relabellings test is the discrete form of unsupervised alignment of similarity structures across individuals by optimal transport [25]. What this paper adds is the measurement in cortex, at the level of single neurons, and the finding that the absolute readout is possible and rests on a remainder that belongs to the correlations, not to the sample of neurons or stimuli.

Code and data. TODO: repository URL. All data are public releases.

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A Scaling and controls

Figures A1 to A4 report the sweeps behind Section 3.5: decoder size, population size, training dynamics and split variance. Two training-side variants were tried and dropped: recomputing each training population’s Gram from a random subset of the stimulus conditions, and zeroing random Gram entries. Neither changes the angular error by more than the seed spread (Section 3.5).

B Controls

Four controls on the MICrONS standard configuration (17M, 512-neuron populations, digital twin unless stated; Table 4).

What the decoder reads. With the three row statistics as the only input per neuron and no relational term, the error is 37.7° (the statistics are not a selectivity confound: their correlations with the distance to the nearest cardinal axis are below 0.12 in magnitude and with gOSI below 0.16, and gOSI is 0.39 for cardinal and 0.37 for oblique neurons); with the statistics plus the Gram entering only as an attention bias, which makes the decoder equivariant to the order of the neurons, it is 19.9° over three seeds, the same as the standard decoder (20.2°), whose input layer also projects the raw Gram row in sample order. The relations carry the readout, and the order-dependent input layer contributes nothing.

Reference decoders that see the activity. To put a scale on what the relations discard, the same transformer was trained with each neuron’s token built from its response vector instead of its Gram row, so that the decoder sees the stimulus-aligned activity that the Gram is computed from (Table 3). It reaches 21.5° in vivo and 15.0° on the twin (three seeds), against 25.4° and 20.2° for the relational decoder, and R^2 0.37 and 0.80 for receptive fields against 0.23 and 0.48. Adding the Gram to this decoder as an attention bias changes nothing (22.2° , 15.0°). A linear readout of each neuron’s own response vector, with no population context at all, does as well as the activity transformer (20.2° , 15.8° ; R^2 0.34, 0.68), so the activity decoder is in effect a per-neuron tuning readout, and what it has over the relational decoder is the alignment of the responses to the stimulus. On orientation the relations retain about 85% of the gain over chance that the aligned activity gives; on receptive fields about 60%; the twin receptive-field gap is inflated by the twin’s labels and responses coming from the same model. What the price buys is independence from the stimulus alignment: with test responses from movie frames the decoder never trained on, the activity decoder has no valid input, its coordinates now index different frames, and it scores 49.5° in vivo and 41.0° on the twin, at or below chance; this is a statement about what a stimulus-aligned readout can and cannot be applied to, not a contest it could have won, whereas the relational decoder under the same test scores 33.8° and 20.7° . The activity decoder transfers across animals under a shared stimulus protocol (Section 3.3) but not across stimuli; the relations do both, because a correlation does not depend on which frames it was computed from, only on both neurons having seen the same ones. Finally, is the Gram all that the relations offer? A decoder given the response vectors with the stimulus bins shuffled, the same shuffle for every neuron of a population, is blind to the stimulus but free to compute any relation among the neurons, pairwise or higher. It reaches 26.3° and 27.8° on orientation over three seeds, no better than the Gram decoder and, on the twin, unstable across seeds (22° to 37°), and R^2 0.04 and 0.08 on receptive fields, far below it; the shuffled vector without population context is near chance (40°), so what the shuffled decoder does read comes from relations it computes itself. We find no evidence that a decoder free to compute richer statistics extracts more than the pairwise Gram, with the caveat that it has to discover the relations under a fresh permutation for every population, a harder problem than being handed them; what the comparison does show is that giving the decoder the relations explicitly is better than leaving it to find them.

Table 3: Reference decoders that see the activity (each neuron’s stimulus-aligned response vector), 17M unless linear, 512-neuron populations. Orientation as mean angular error (chance 45°), receptive field as R^2 ; mean $\pm$ s.d. over three seeds where given, otherwise seed 0.

decoder input	ori. in vivo	ori. twin	RF in vivo	RF twin
Gram row (this paper)	$25.4^\circ \pm 0.5^\circ$	$20.2^\circ \pm 0.4^\circ$	0.23 ± 0.01	0.48 ± 0.01
activity, transformer	$21.5^\circ \pm 0.4^\circ$	$15.0^\circ \pm 0.1^\circ$	0.37	0.80
activity + Gram as attention bias	22.2°	15.0°	–	–
activity, linear, no population context	$20.2^\circ \pm 0.2^\circ$	$15.8^\circ \pm 0.1^\circ$	0.34 ± 0.01	0.68 ± 0.01
activity, transformer, disjoint stimulus bins	49.5°	41.0°	–	–
activity, bins shuffled per population, transformer	$26.3^\circ \pm 0.6^\circ$	$27.8^\circ \pm 6.5^\circ$	0.04	0.08
activity, bins shuffled, linear, no population context	40.3°	40.2°	–	–

Cross-stimulus test. Training and selection Grams built from one random half of the stimulus bins and test Grams from the other half give 20.7° on the twin (2,500 bins per half, raw responses rather than principal components; three seeds, 20.3 to 21.2) against 20.2° with shared bins, and 33.8° in vivo (60 bins per half; 33.4 to 34.2) against 25.4° . A control with the same 60-bin half on both sides gives 25.3° , so the in-vivo cost is not the smaller Gram but the stimulus sample: a correlation matrix over 60 movie bins partly reflects which frames it was built from, and one over 2,500 bins does not. The relational structure that the decoder reads is stable across stimuli once the stimulus sample is large.

Scan and area. The MICrONS neurons come from 13 scans in four areas, same-scan pairs correlate more strongly than cross-scan pairs, and the scans look at different parts of the screen, so a decoder that read scan identity from the Gram could predict each scan’s mean receptive-field position and score a positive R^2 without knowing any neuron’s own position. That is what the Allen decoder does (Section 3.4), so we checked MICrONS the same way, centring predictions and labels per scan (and per area) before scoring. Receptive fields survive: R^2 is 0.50 absolute and 0.50 within scan on the twin, 0.25 and 0.25 in vivo, and 0.47 and 0.22 within area. Only 2% of the label variance lies between scans and 9% between areas, so a scan-identity readout could reach at most 0.02. For orientation, a predictor that assigns every neuron its scan’s or area’s mean orientation scores 39° , and removing a separate rotation per scan or area changes the decoder’s error by less than 0.2° . Both MICrONS results are neuron-level readouts within a scan.

Class balance and priors. Preferred orientations are unevenly distributed in MICrONS (19% of labelled neurons within 11° of 0° , 25% of 90° , 6 to 10% in each oblique bin), so a decoder could gain from the distribution alone. It gains little: the best constant prediction has an error of 41.3° , a predictor that assigns every neuron the circular mean of its scan 39.0° and of its area 38.7° , against 20.3° for the decoder on the twin and 24.6° in vivo (neuron split 2). Averaging the error over eight equally weighted true-orientation bins gives 24.5° on the twin and 27.9° in vivo. Training and testing on populations drawn with equal numbers of neurons from each of the eight classes (64 per class) gives 22.0° on the twin and 26.6° in vivo (two seeds: 22.4, 21.5; 26.5, 26.7), raw and after global alignment alike, so the absolute frame does not depend on the class imbalance. Removing a separate rotation per scan or per area changes the error by less than 0.2° , and the per-area errors are within 3° of each other. On Allen the labels are nearly flat (14 to 20% per orientation), the best constant is 43.9° , and the balanced error is 36.4° against 35.7° raw. What the per-bin errors do show, on both datasets, is that the readout is cardinal: neurons preferring 0° or 90° are decoded to 11° to 12° on the twin and 25° on Allen, oblique ones to 37° to 45° , and 63 to 72% of all predictions fall on the two cardinal axes (Fig. A5). The relations say how close a neuron is to a cardinal axis, which is the anisotropy of Section 3.2 seen from the decoder’s side, and much less about which oblique it prefers.

Table 4: Controls, mean angular error on held-out neurons (chance 45°). Where two numbers are given, the second is the error after the best global rotation or reflection (rows marked *aligned*) or averaged over eight equally weighted true-orientation bins (rows marked *balanced*).

control	substrate	error
standard decoder	twin	20.2°
row statistics only, no relational term	twin	37.7°
row statistics + Gram as attention bias (3 seeds)	twin	$19.9^\circ \pm 0.3^\circ$
disjoint stimulus bins, training vs test (3 seeds)	twin	$20.7^\circ \pm 0.4^\circ$
standard decoder	in vivo	25.4°
disjoint stimulus bins (60 per side; 3 seeds)	in vivo	$33.8^\circ \pm 0.3^\circ$
same 60 bins on both sides	in vivo	25.3°
neuron split 1 / 2	in vivo	25.4° / 24.6°
standard decoder, raw / balanced	twin	20.3° / 24.5°
orientation-balanced loss, raw / balanced	twin	21.4° / 25.3°
standard decoder, raw / balanced	in vivo	24.6° / 27.9°
orientation-balanced loss, raw / balanced	in vivo	25.8° / 29.1°
class-balanced populations (64 per class), raw / aligned, 2 seeds	twin	22.0° / 21.9°
class-balanced populations, raw / aligned, 2 seeds	in vivo	26.6° / 26.4°
trained on V1, scored on V1, raw / aligned	twin	22.2° / 22.2°
trained on V1, scored on RL, raw / aligned	twin	26.2° / 24.6°
trained on V1, scored on AL, raw / aligned	twin	58.2° / 31.6°
within-scan populations (128 neurons)	twin / in vivo	30.7° / 32.6°
trained on one scan (140 neurons), tested on the rest	twin / in vivo	38° – 44° / 37° – 39°
mixed-scan populations (128 neurons, V1)	twin	22.2°

Synthetic populations. 6,000 model neurons with von Mises tuning ($\kappa = 2$), preferred orientations either uniform or with half the neurons within about 12° of a cardinal axis (count), tuning either isotropic or sharpened to $\kappa = 4$ at the cardinals (sharp), and a 120-frame stimulus that either covers orientation evenly or has half its frames near a cardinal axis (stim); responses with Gaussian noise; the 2.2M decoder on 512-neuron populations, epoch selected on the raw error as in every other run, five data and decoder seeds (Table 5). With every switch off the class-Gram is circulant to within noise (neighbour-correlation range 0.01 against 0.04 to 0.14 with a switch on), and the decoder recovers the ordering but no frame: 30° after one global alignment, 54° raw. Sharper cardinal tuning and a cardinal excess of neurons do not fix the frame, alone, together, or at any strength (κ up to 32; gain up to 5 times; two seeds each, all above 42° raw); stimulus anisotropy fixes it in most seeds. The anisotropy the cortical data show, stronger co-fluctuation among neurons that prefer the area’s dominant orientation (Section 3.2), is modelled by a smooth random field over preferred orientation entering each neuron with a weight peaked at 90° , with uniform counts, isotropic tuning and an even stimulus. Scaled to raise the same-class correlation at 90° by a factor of 1.55, it fixes the frame in four of five seeds at 9° raw; at 2.1 in both seeds at 3° ; at 1.2, the factor of the twin V1 data, in one of two. The spread across seeds is large in every cell where the frame is not fixed, because selection on the raw error then has no signal to select on; the cell means are reported with their range for that reason.

C Classification results

Earlier runs decoded orientation as a class (8 classes on MICrONS, 6 on Allen) with the same decoder and protocol; the per-neuron accuracies are in Table 6. They lead to the same conclusions, but a class score counts a near miss as a miss, so the angular error in the main text is the better summary of a continuous label.

Table 5: Synthetic populations, five seeds: raw error / error after the best global rotation or reflection, mean over seeds with the range in brackets, and the number of seeds in which the frame was fixed (raw error below 30° and within 5° of the aligned error). Bottom block: shared variability among like-tuned neurons entering with a weight peaked at 90° , scaled so that the same-class correlation at 90° rises by the stated factor. A gap between raw and aligned error means structure without frame.

anisotropy switched on	raw	aligned	frame fixed
none (circulant)	53.7° [44–63]	30.0° [23–42]	0 / 5
stimulus	23.5° [4–54]	13.9° [4–19]	3 / 5
sharper cardinal tuning	51.4° [39–65]	28.6° [15–42]	0 / 5
cardinal excess of neurons (count)	37.8° [27–45]	35.1° [27–42]	1 / 5
sharper tuning + stimulus	12.2° [3–38]	6.3° [3–10]	4 / 5
count + stimulus	25.0° [4–47]	20.7° [4–43]	3 / 5
count + sharper tuning	42.8° [29–58]	35.2° [27–41]	1 / 5
all three	18.4° [3–46]	17.9° [3–44]	4 / 5
like-tuned co-fluctuation at 90° , $\times 1.55$	8.9° [4–26]	7.3° [3–19]	4 / 5
$\times 1.2$ (two seeds)	31.5° [15–48]	24.2° [15–33]	1 / 2
$\times 2.1$ (two seeds)	2.6° [2–3]	2.5° [2–3]	2 / 2
$\times 1.55$ + stimulus (two seeds)	5.2° [4–6]	5.2° [4–6]	2 / 2

Table 6: Orientation as classification, 17M decoder. MICrONS: 8 classes, majority class 0.255, mean $\pm$ s.d. over 3 seeds. Allen: 6 classes, majority class 0.19 to 0.20, three splits per cell (17M decoder with the condition-subset augmentation, 256-neuron populations; the `pooledcross` split-0 value is the mean of 3 seeds).

dataset	cell	accuracy
MICrONS	in vivo	0.380 ± 0.006
MICrONS	digital twin	0.420 ± 0.003
Allen	training animals, single-animal populations	0.210 / 0.216 / 0.217
Allen	training animals, mixed populations	0.281 / 0.283 / 0.283
Allen	held-out animals, single-animal populations	0.199 / 0.197 / 0.187
Allen	held-out animals, mixed populations	0.293 / 0.236 / 0.264

D Labelled reference models

A second reference uses labels at test time: a ridge readout from each test neuron’s correlations to the labelled training neurons, fitted on the training half, from the classification-era runs (orientation as 8 or 6 classes, accuracy; receptive fields as R^2). It is one linear labelled model, not a ceiling, and the relational decoder exceeded it on Allen receptive fields.

Table 7: Ridge readout from a neuron’s correlations to labelled training neurons (classification era; orientation accuracy, receptive field R^2).

dataset, substrate	orientation	receptive field
MICrONS, in vivo	0.41 (8 classes)	0.32
MICrONS, digital twin	0.49 (8 classes)	0.67
Allen, leave-one-mouse-out	0.35 (6 classes)	0.20

E Receptive-field decomposition per split

Per held-out-mouse split, for the two Allen regimes of Fig. 6: R^2 on absolute screen coordinates, the correlation between predicted and true per-mouse mean position over the 9 test mice (x, y), and the correlation between predicted and true position relative to the mouse mean over the cells (x, y).

Table 8: Allen receptive fields per split.

populations	split	absolute R^2	per-mouse mean $r(x, y)$	within-mouse $r(x, y)$
single-animal	0	0.29	0.47, 0.74	0.07, 0.11
single-animal	1	−0.02	0.48, 0.40	0.06, 0.08
single-animal	2	−0.45	0.19, 0.25	0.11, −0.01
mixed	0	0.22	0.70, 0.81	0.00, 0.20
mixed	1	0.08	0.23, 0.80	0.02, 0.08
mixed	2	−0.25	0.46, 0.75	−0.06, 0.04

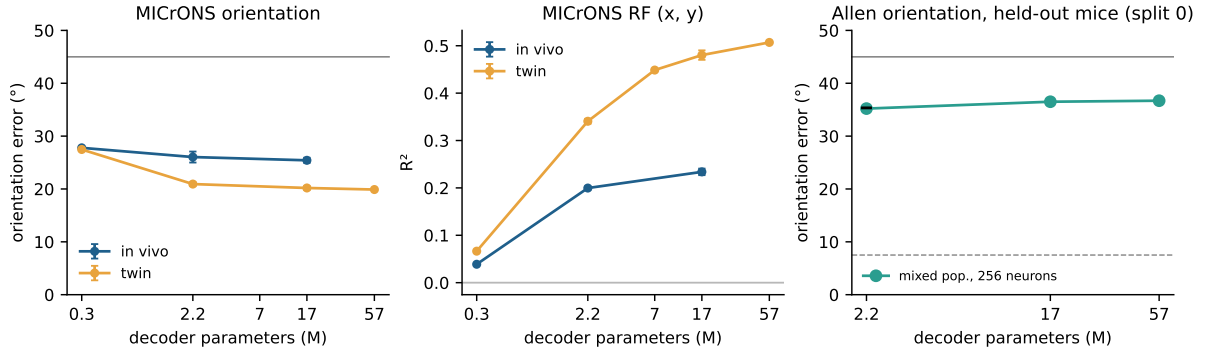


Figure A1: Scaling with decoder size. Test metric versus number of decoder parameters at fixed population size (512 neurons MICrONS, 256 Allen). Error bars: s.d. over 3 seeds where run. On MICrONS receptive fields the gain continues to 57M; the orientation error is flat from 2.2M upward on both substrates and on Allen.

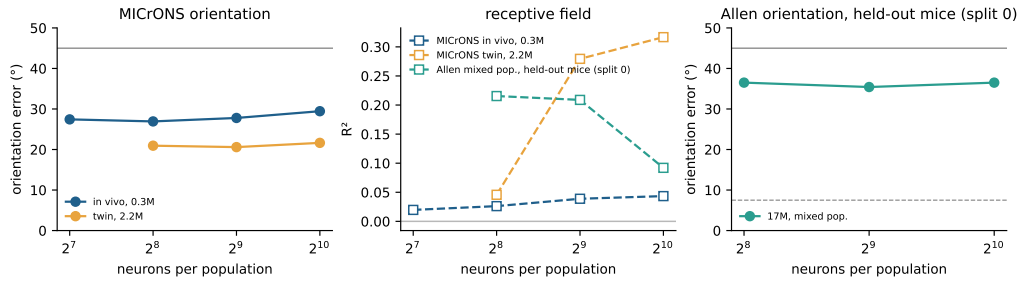


Figure A2: Population size. Test metric versus the number of neurons in each sampled population. Beyond 256 to 512 neurons the orientation error moves by at most 2° , up or down; twin receptive fields keep gaining slowly, Allen receptive fields lose.

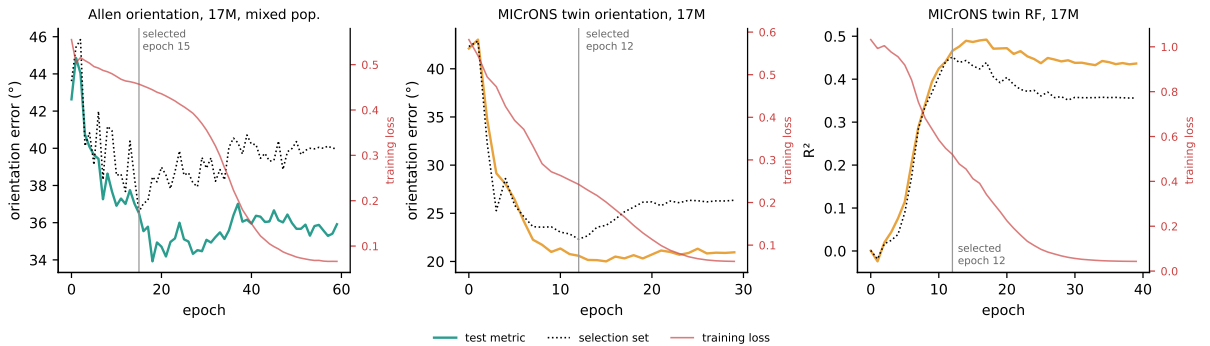


Figure A3: Training dynamics and epoch selection. Test metric (solid), selection-set metric (dotted) and training loss (red, right axis) per epoch. The selection set, never the test set, picks the epoch (grey line). The training loss collapses within 10 to 20 epochs on every large run.

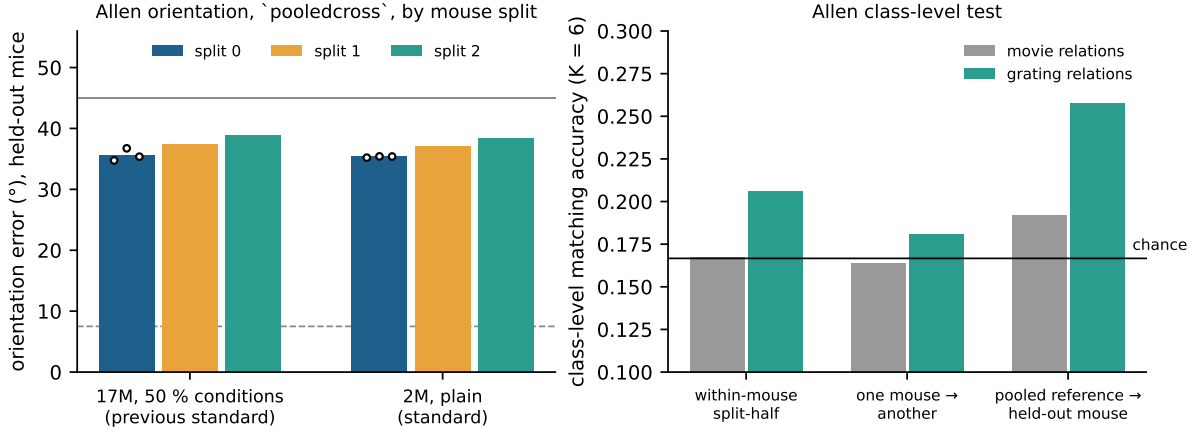


Figure A4: Allen split variance and the class-level test. Left: `pooledcross` orientation error on the same three held-out-mouse splits for the previous standard configuration (17M, augmented) and the final one (2.2M, plain); split 0 carries three seeds. Right: the class-level test with relations from natural movies versus drifting gratings; only the grating relations move every test off chance.

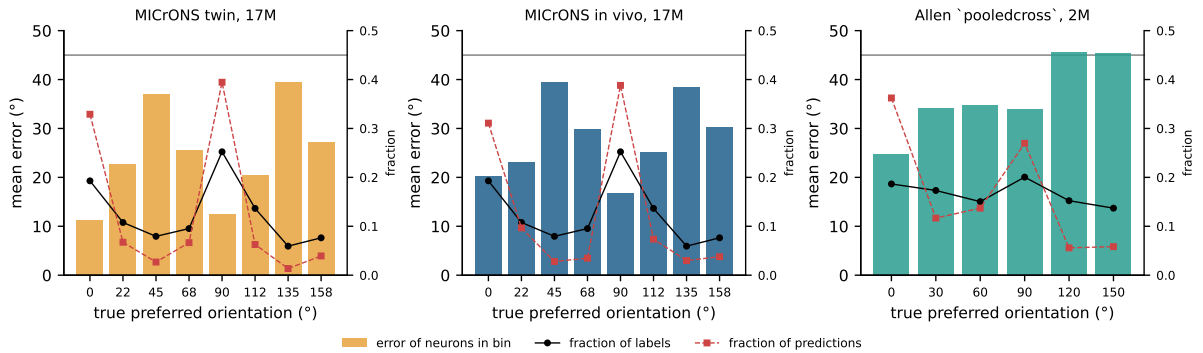


Figure A5: The readout is cardinal. Mean angular error of the neurons in each true-orientation bin (bars; grey line: 45° chance), with the fraction of labels (black) and of predictions (red) per bin, for the twin and in-vivo MICrONS decoders (neuron split 2) and the Allen `pooledcross` decoder. Cardinal orientations are decoded well, obliques near chance, and most predictions fall on 0° or 90°.